FORMAL DEVIATION & OUT-OF-SPECIFICATION (OOS) INVESTIGATION REPORT

Document ID: DEV-2026-0814 | QMS Reference: QMS-OOS-26-042 | Logged Date: 2026-07-14

1. BATCH & ACTIVE PHARMACEUTICAL INGREDIENT (API) IDENTIFICATION

Product Name:	Metformin Hydrochloride API	Batch / Lot Number:	MFH260712A
Pharmacopoeial Grade:	USP / Ph. Eur. / IP Grade	Manufacturing Date:	2026-07-12
Manufacturing Facility:	API Block-3, Synthesis Suite B	Expiry / Retest Date:	2031-07-11 (60 Months)
Batch Quantity Affected:	150 kg (6 HDPE Drums, 25 kg/drum)	Packaging Configuration:	Double Polyethylene (LDPE) Liners
Dedicated Equipment Train:	Reactor R-301, Centrifuge CF-201, Fluidized Bed Dryer FBD-301	Investigation Category:	OOS - QC Assay Out of Specification
Detection Method:	QC Release Testing (HPLC Method STP-MET-004)	Initial Severity Classification:	Major Process Deviation

2. INCIDENT DESCRIPTION & OUT-OF-SPECIFICATION FINDINGS

During routine QC release testing of Metformin Hydrochloride API batch MFH260712A, the QC analytical laboratory reported an assay result of 97.20% w/w (on dried basis), which fails the approved USP/Ph. Eur. release specification of 98.0% to 102.0% w/w. The initial duplicate analysis was performed on HPLC instrument HPLC-08 per standard test procedure STP-QC-MET-004.

Duplicate injection re-test yielded 97.18% w/w, confirming the Out-of-Specification (OOS) condition. All other chemical parameters tested complied with release limits: Related Substance Impurity B (Dicyandiamide) was detected at 0.08% (limit NMT 0.10%), Loss on Drying was 0.32% (limit NMT 0.50%), and heavy metals were within USP limits. Because this batch fails the primary assay release specification, it cannot be certified for commercial release and was placed under immediate QA quarantine lock.

3. IMMEDIATE CONTAINMENT ACTIONS & QUARANTINE STATUS

- Physical Quarantine: All 6 HDPE drums (150 kg net weight) of batch MFH260712A were affixed with fluorescent red QA Quarantine Hold labels and immediately transferred to Secure Quarantine Room Q-2 under dual padlock authorization.
- ERP System Block: The SAP Material Management status was locked to Blocked Stock (Status 'S'), preventing any warehouse transfer, customer reservation, or shipping authorization.
- Equipment Lockout: Fluidized Bed Dryer FBD-301 was tagged out of service with a Red Maintenance Tag pending instrumentation and temperature sensor calibration verification.
- Traceback & Adjoining Batches: Preceding batch MFH260711Z and succeeding batch MFH260713B were inspected; process batch records confirm no physical crossover or equipment contamination occurred.

4. INITIAL GxP RISK EVALUATION

Classification: MAJOR RISK (Risk Priority Score: 15 / 25).
Patient Impact: Nil currently, as the batch is completely isolated in quarantine before commercial release. If released sub-potent, reduced therapeutic efficacy could occur in type-2 diabetes management.
Regulatory & Compliance Impact: High. An OOS investigation must follow FDA 21 CFR 211.192 and EU GMP Annex 16 guidelines. Investigation SLA is 14 calendar days with mandated QP sign-off.

5. PROCESS OPERATIONAL FORENSICS & SCADA TREND AUDIT

A rigorous audit of the manufacturing batch record (BMR # BMR-MET-2026-088) and SCADA automation trends revealed two critical operational deviations during Stage-4 manufacturing:

1. Fluid Bed Drying Thermal Excursion: During drying in FBD-301 (12-Jul-2026, 14:10 to 14:38 IST), product thermocouple TE-302 logged an anomalous temperature excursion spiking from target setpoint 65°C to 74.2°C for 28 minutes. Operator log notes that chilled water valve CV-102 pneumatic actuator had a delayed response, causing localized overheating.

2. Mother Liquor Secondary Wash Loss: Mother liquor assay analysis from Centrifuge CF-201 filtrate showed a 2.8% API loss during the post-crystallization solvent wash, indicating incomplete crystallization before spinning caused by inadequate cooling bath stabilization (crystallizer remained at 22°C instead of standard 16-18°C setpoint).

6. QC LABORATORY PHASE I TESTING & SYSTEM SUITABILITY

- Analyst Verification: The testing chemist was fully qualified and certified on analytical method STP-QC-MET-004.
- HPLC System Performance: System suitability parameters passed (Retention time: 6.42 min, USP theoretical plates: 3,840 [criterion: >2,500], tailing factor: 1.18 [criterion: <1.5], RSD of 6 replicate injections of standard: 0.42% [criterion: <1.0%]).
- Reference Standard & Glassware: Metformin standard lot RS-MET-2025-09 was within valid expiry. Grade-A calibrated volumetric flasks and pipettes were verified.
- Conclusion: No laboratory testing calculation or dilution error was detected; the OOS is attributed to a genuine manufacturing process variation.

7. CRITICAL MISSING EVIDENCE GAPS IDENTIFIED FOR ROOT CAUSE CLOSURE

To achieve defensible regulatory closure, the Lead Investigator must collect and verify the following 4 evidence items:

1. Calibration and qualification records for the HPLC system and analytical balance BAL-042 on the testing day (14-Jul-2026).
2. Fluidized bed dryer FBD-301 thermocouple TE-302 temperature trend logs and 6-month calibration log sheet.
3. Raw material vendor Certificate of Analysis (CoA) and inbound quarantine batch release verification for Dimethylamine HCl.
4. Preventative maintenance records and pneumatic seal integrity inspection for chilled water valve CV-102.

8. INVESTIGATION TEAM ASSIGNMENT & GxP APPROVAL SIGN-OFF

Lead Investigator: Dr. Arvind Sharma (Senior QA Specialist)

Prescribed Investigation Methods: 5 Whys Root Cause Analysis and Ishikawa 6M Fishbone Diagram

Mandated Target Closure Date: 2026-07-28 (14-Calendar-Day SLA)

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QC Analyst | 14-Jul-2026	Principal QA Engineer | 14-Jul-2026	Qualified Person (QP) | 14-Jul-2026